Bridge Day 6 STUDENT QUIZ

Multi-Step Calculations, Summaries, and Conditional Output

Monday, July 13, 2026 • 20 points • target 17 minutes

Name		UIN		Section	

Assessment boundary and evidence source

Contract: 07a04826c975
Cumulative foundations: input conversion, arithmetic, min/max/sum/len, Boolean decisions, f-strings
Scaffold level: complete geometry and classification program
Evidence source: the matching v5 workbook readiness/evidence pages and VIP activities 18.8, 18.9, 18.10, 18.11, 18.12.
Show intermediate state for trace credit. Program credit requires one coherent solution, a stated assumption when the prompt leaves one implicit, and at least one test whose expected result can distinguish correct from incorrect logic.

Item 1. Evaluate ceiling-style kit arithmetic. - 4 points

```
groups = 7
items_per_group = 5
spares = 4
items = groups * items_per_group + spares
kits = (items + 11) // 12
leftover_space = kits * 12 - items
print(kits, leftover_space)
```

Question: Show the item total, adjusted numerator, quotient, and exact output.

Item 2. Trace a numeric summary and formatting boundary. - 4 points

```
values = [4.5, 7.0, 2.5, 6.0]
total = sum(values)
average = total / len(values)
print(min(values), max(values), f"{average:.2f}")
```

Question: Compute every summary and give exact output, including the formatted average.

Item 3. Combine arithmetic with a conditional label. - 4 points

```
mass = 18.0
purity = 84.5
cost = 12 + 1.8 * mass
hold = purity < 85 or mass > 25

if hold:
    label = "hold"
else:
    label = "accepted"

print(f"{label}:  {cost:.2f}")
```

Question: Show the cost calculation and each Boolean component before giving exact output.

Complete program - 8 points

- Read length, width, and height as floats.
- Print invalid if any dimension is not positive.
- Otherwise compute volume and rectangular-prism surface area.
- Classify oversize when volume is above 500 or the largest dimension is above 12; otherwise classify standard.
- Print Volume and Surface area to two decimals, followed by Class: and the label.

Program workspace

Test input, expected result, and why this test matters

Quiz evidence is sampled from the taught construct boundary; workbook and notebook evidence remain the primary learning surfaces.